



Academy Lab Projects – Showcase

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Cloud Web Application Builder

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Business scenario overview

Problem / Opportunity

- ✓ Example University's admissions web application often slows down or becomes inaccessible during peak admission periods.
- ✓ Thousands of students need to submit and check applications simultaneously, but the system cannot handle the load.
- ✓ Lack of scalability leads to poor user experience, system downtime, and complaints from students.

Business scenario overview

Solution Requirements

- ✓ Must support thousands of concurrent users.
- ✓ Functional CRUD operations: View, Add, Delete, Modify student records.
- ✓ Load balanced across multiple servers.
- ✓ Highly available (minimal downtime).
- ✓ Database securely isolated (no direct public access).
- ✓ Secure communication between components using appropriate ports.
- ✓ Scalable via AWS Auto Scaling.
- ✓ Cost-optimized while maintaining high performance.

Solution overview

- ✓ The project involves designing and deploying a proof-of-concept student records web application on AWS Cloud.
- ✓ It uses a **multi-tier architecture** consisting of:
- ✓ **Web Layer**: Amazon EC2 instances hosting the application code.
- ✓ **Database Layer**: Amazon RDS (MySQL).
- ✓ **Networking Layer**: Custom VPC with public/private subnets.
- ✓ **Security Layer**: Security Groups, IAM roles, AWS Secrets Manager.
- ✓ **Scalability/Availability Layer**: Application Load Balancer and Auto Scaling Group.

Design Considerations

- ✓ Separate application and database tiers for security and scalability.
- ✓ Private subnets for databases to prevent external attacks.
- ✓ Auto Scaling with minimum/maximum instance thresholds.
- ✓ Cost efficiency by starting with t3.micro instances.
- ✓ Use of AWS Cloud9 and AWS CLI for automation.

Use Cases

- ✓ Students: Access and update application data online without delays.
- ✓ Admissions staff: Manage and verify student records in real-time.
- ✓ University IT Team: Scale resources automatically during peak loads and minimize cost during low activity.

Architecture diagram of the solution

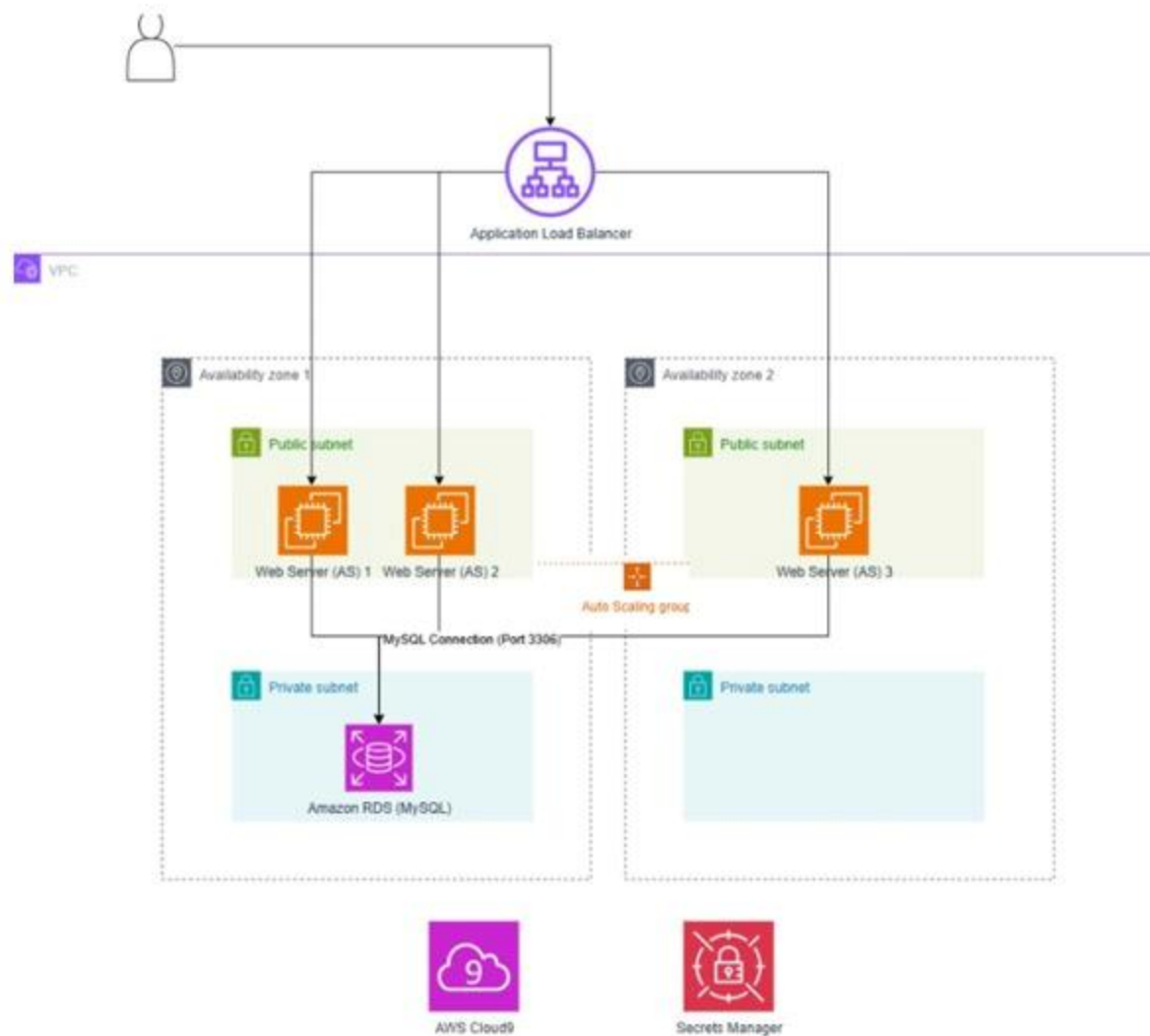


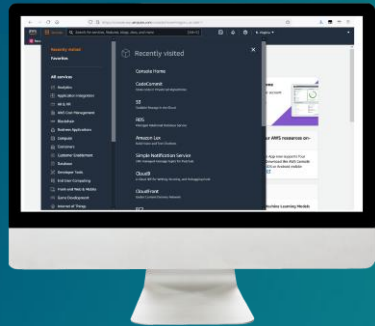
Figure 1:

Architecture diagram of the solution

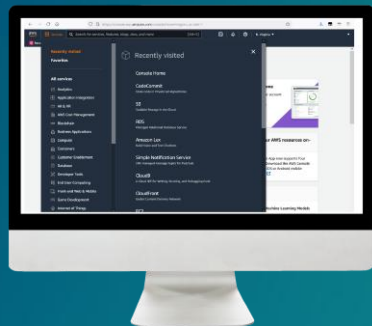
Figure 1: Final High-Availability Architecture

- Application Load Balancer distributes traffic across EC2 web servers.
- Web servers deployed across **two Availability Zones** for resilience.
- Amazon RDS (MySQL) hosted in private subnet (secure, no public access).
- Auto Scaling group ensures scalability.
- AWS Cloud9 supports deployment and testing.
- Secrets Manager manages DB credentials securely.

Claimed Badge



Project Screenshots



aws ec2

VPC > Your VPCs

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

- Your VPCs**
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers [New](#)

Security

- Network ACLs
- Security groups

Your VPCs (2) Info

Find VPCs by attribute or tag

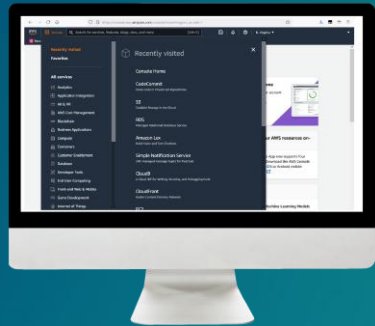
☐	Name	VPC ID	State	Block Public...	IPv4 CIDR
☐	it21192364vpc-vpc	vpc-01a099b79c1dd012f	Available	Off	10.0.0.0/24
☐	-	vpc-019ad0a46b4d87325	Available	Off	172.31.0.0/16

Select a VPC above

CloudShell Feedback

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Project Screenshots



aws | cloud9

EC2 > Instances

Instances (1/2) Info

Last updated 23 minutes ago

Connect Instance state All states

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	it21192364one	i-0e14c5840b83efe13	Running	t2.micro	Initializing	View alarms +
☐	aws-cloud9-it...	i-062fdd43198677452	Running	t3.micro	Initializing	View alarms +

i-0e14c5840b83efe13 (it21192364one)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID
i-0e14c5840b83efe13

IPv6 address
-

Public IPv4 address
3.87.93.221 | open address

Instance state
Running

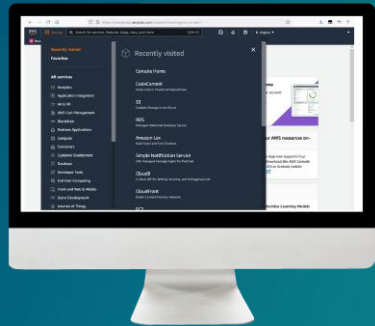
Private IPv4 addresses
10.0.0.4

Public DNS
ec2-3-87-93-221.comp
address

CloudShell Feedback

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Project Screenshots



← → ↻ ⚠ Not secure 3.87.93.221/students



XYZ University

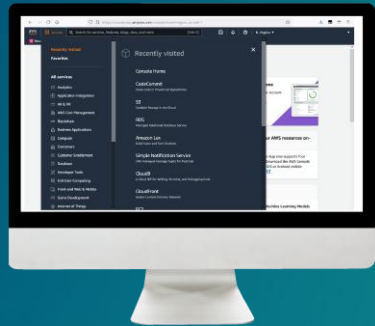
[Home](#)
[Students list](#)

All students

Name	Address	City	State	Email	Phone
Deshan	Malabe	Malabe	West	deshan123@gmail.com	0756645624

[Add a new student](#)

Project Screenshots



aws | Q rds

United States (N. Virginia)

AWS Cloud9 > Environments

AWS Cloud9

My environments
Shared with me
All account environments

Documentation

Successfully created it21192364cloud9. To get the most out of your environment, see [Best practices for using AWS Cloud9](#).

For capabilities similar to AWS Cloud9, explore AWS Toolkits in your own IDE and AWS CloudShell in the AWS Management Console. [Learn more](#)

Environments (1) Delete View details Open in CloudShell

My environments

	Name	Cloud9 IDE	Environment type	Connection	Permission	Owner ARN
☐	it21192364cloud9	Open	EC2 instance	Secure Shell (SSH)	Owner	arn:aws:sts::271828048423:role/voclabs/user4457521=IT21192364cloud9

CloudShell Feedback

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Lessons learned

Challenges Overcome

- Migrating DB from EC2 → RDS.
- Configuring VPC with both public and private subnets.
- IAM and Secrets Manager integration.
- Auto Scaling configuration.

• Helpful Resources

- AWS Academy labs.
- AWS Documentation + Architecture diagrams.
- AWS Pricing Calculator.

Lessons learned

New Skills Gained

- Multi-tier architecture deployment.
- Auto Scaling + ALB integration.
- Secure DB connection via Secrets Manager.
- Load testing with Cloud9.

Next Steps

- Enable HTTPS with SSL (ACM).
- Add CloudWatch monitoring & alarms.
- Implement Multi-AZ RDS for failover.
- Explore multi-region deployments for disaster recovery.



Thank you